Principal Component Analysis

Mathematical Foundations of Machine Learning

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Why reduce dimensions?



300 × 300 pixels

251 244 239	212 192 179	109 25 9	146 57 7	124 53 20
251 242 237	248 241 234	126 52 32	128 40 7	113 48 15
251 244 237	252 247 241	172 125 105	122 31 4	101 39 13
251 242 234	250 242 232	216 188 171	126 39 1	95 36 12
251 244 237	251 244 238	234 218 205	129 40 2	93 33 12

each pixel = 3 numbers (R, G, B)

300 × 300 × 3

= 270,000

numbers

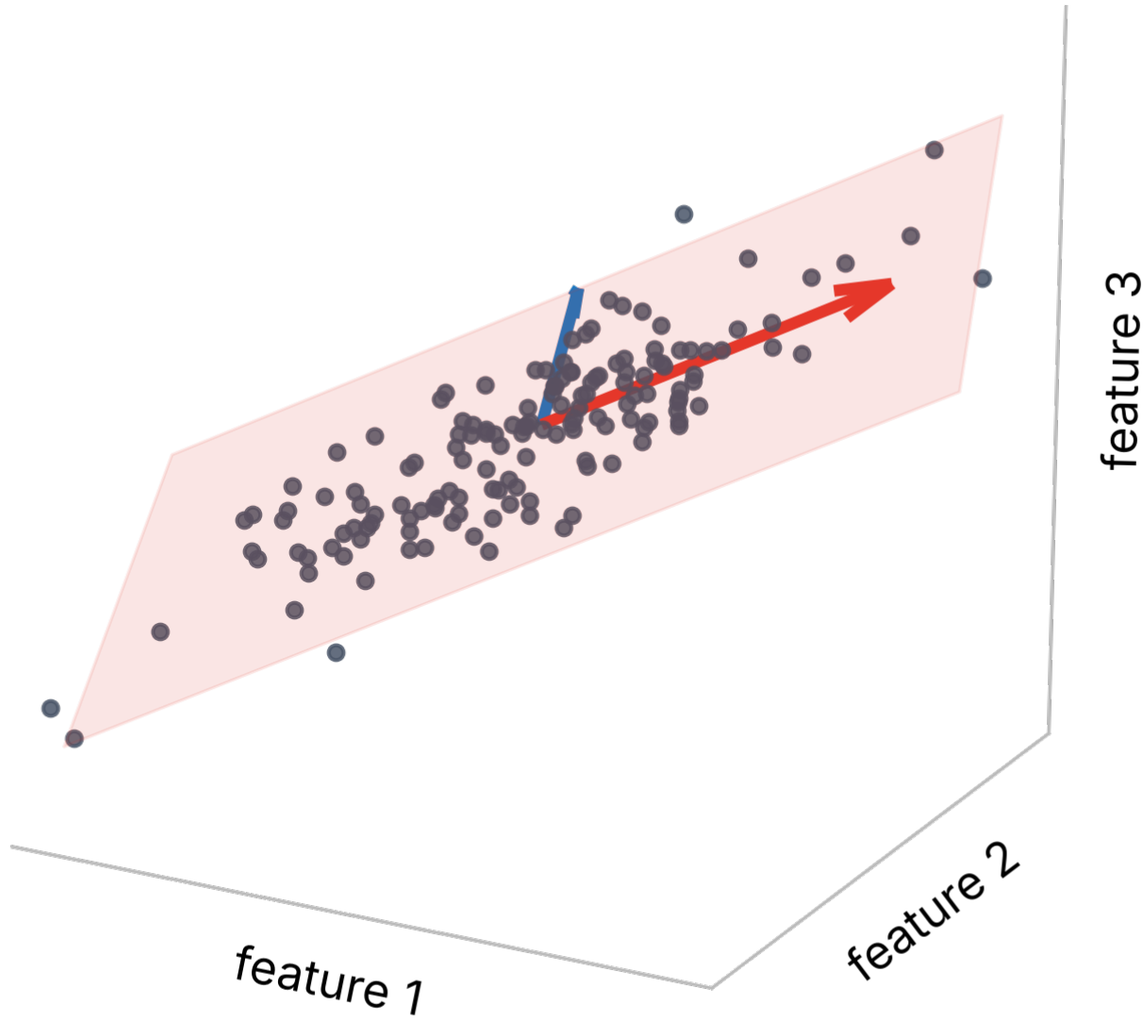
one image = one point in $\mathbb{R}^{270,000}$

Challenges of High Dimensional Data

- Hard to analyze
- Difficult to interpret and visualize
- Computationally expensive

Good News

Data may lie near a lower-dimensional subspace.



3 coordinates per point, but the data is basically a flat sheet

Mean

Mean denoted by μ is the average value in a collection of numbers.

$$\mu = \frac{1}{m}\sum_{i=1}^{m} x_i$$

$$X = \{3,7,5\}$$

$$\mu = \frac{3+7+5}{3} = 5$$

Variance

Variance denoted by σ^2 is a statistical measurement of the spread between the numbers in a data set.

$$\sigma^2 = \frac{1}{m-1}\sum_{i=1}^{m}(x_i - \mu)^2$$

Example

$$X = \{3,7,5\}, \quad \mu = \frac{3+7+5}{3} = 5$$

$$\sigma^2 = \frac{1}{2}((3-5)^2 + (7-5)^2 + (5-5)^2) = \frac{8}{2} = 4$$

Covariance

Covariance is a statistical measure of the strength and sign of the linear relationship between two variables in the scale of the original data.

$$Cov[X, Y] = \frac{1}{m-1} \sum_{i=1}^{m} [(x_i - \mu_x)(y_i - \mu_y)]$$

$$Cov[X, X] = Var[X]$$

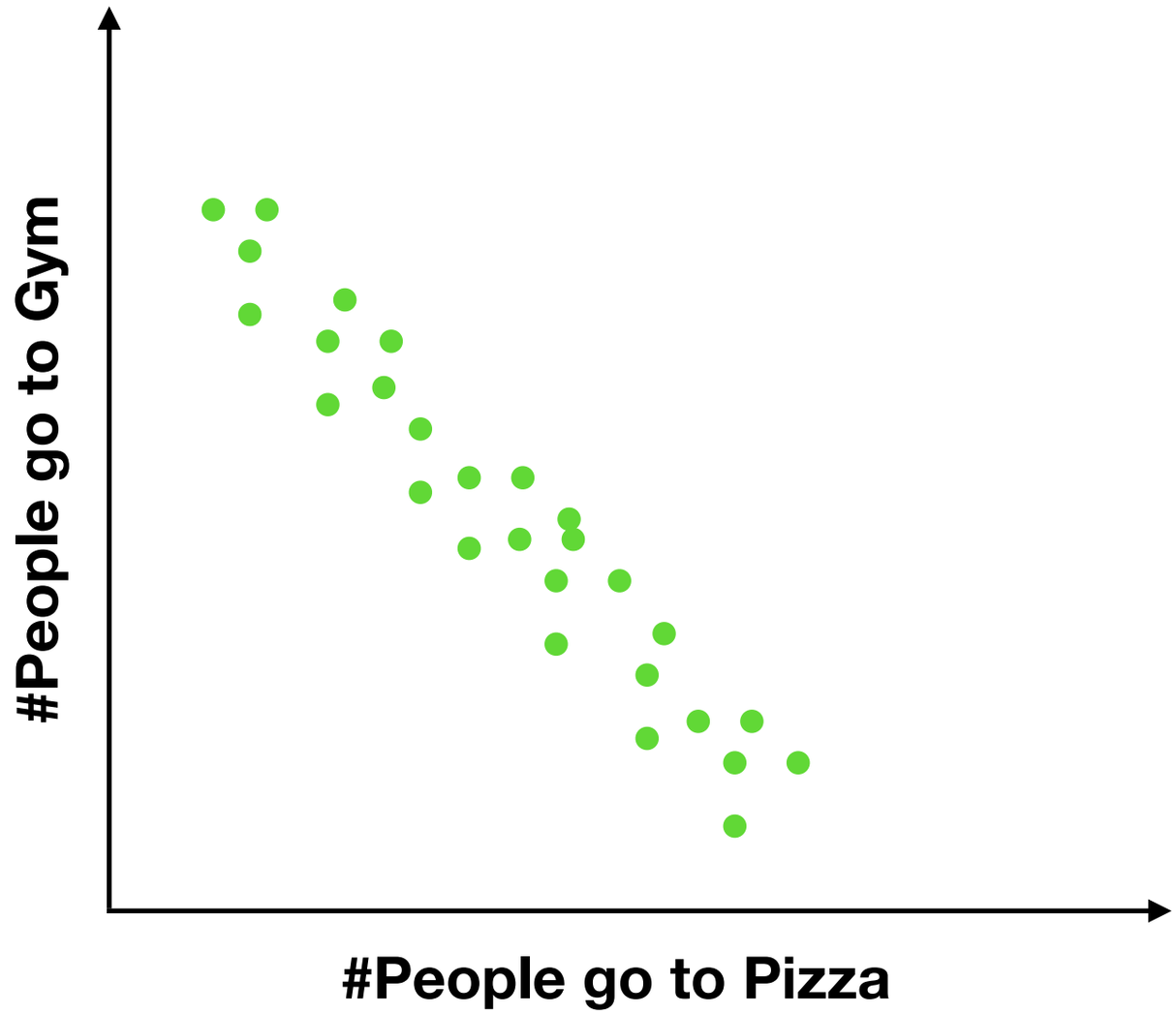
Covariance Matrix

$$\begin{bmatrix} Var[X] & Cov[X,Y] \\ Cov[Y,X] & Var[Y] \end{bmatrix}$$

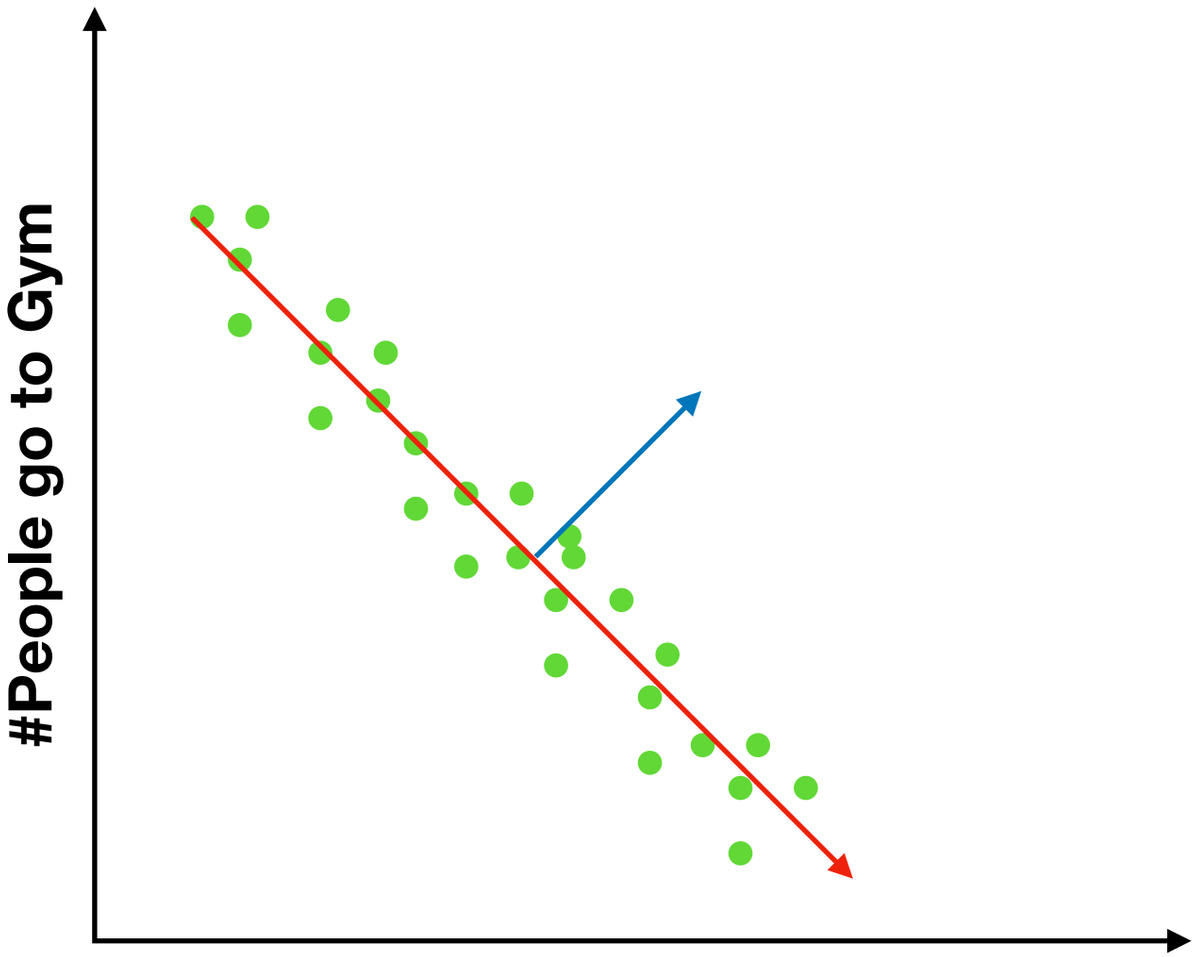




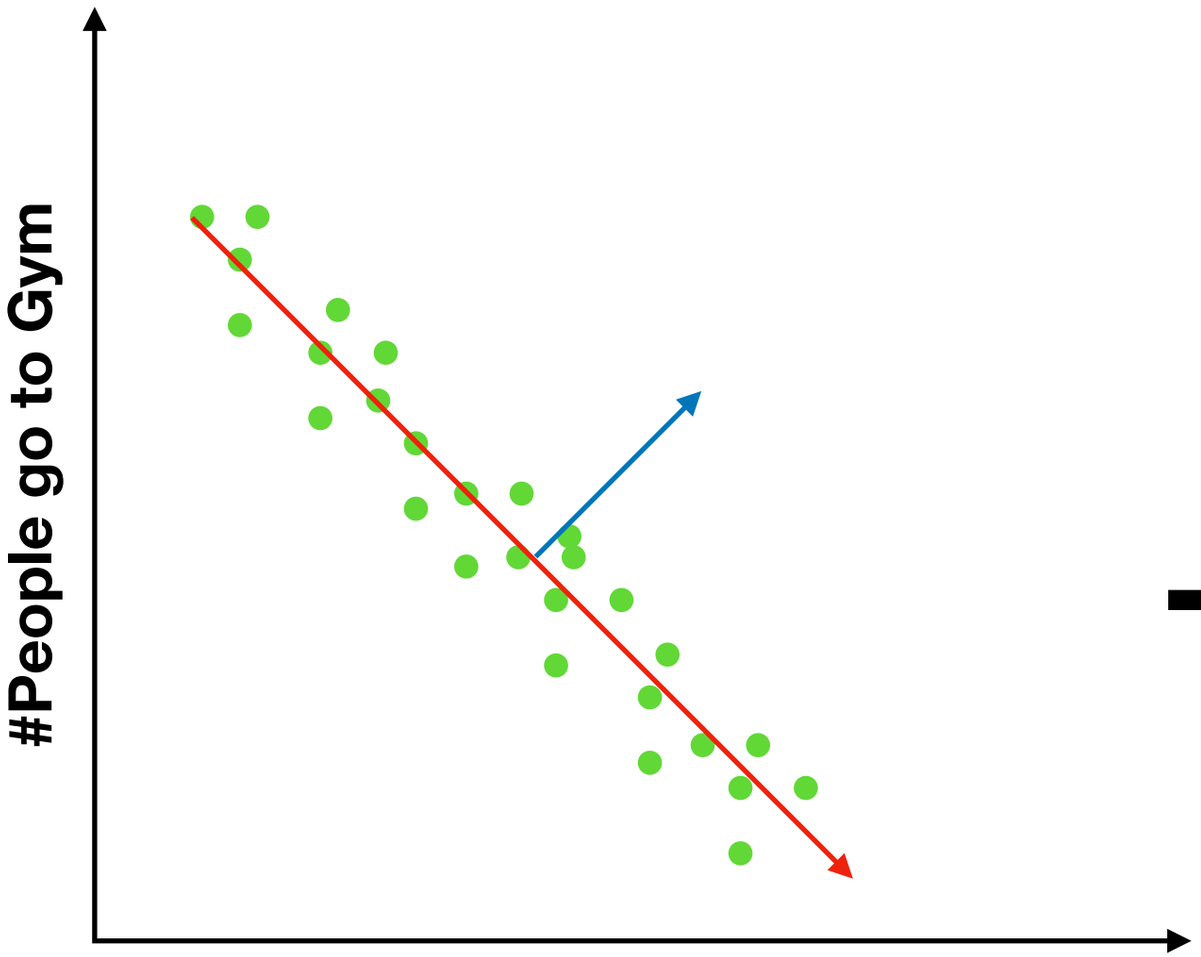
@Freepik

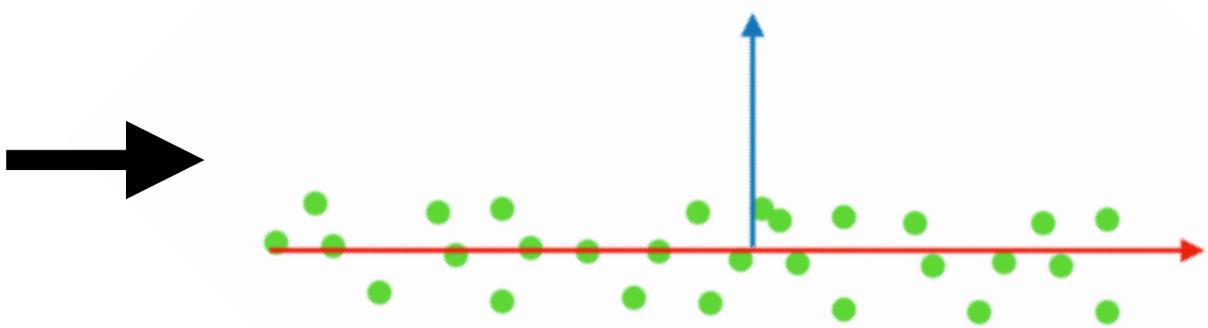

#People go to Gym
#People go to Pizza


#People go to Gym
#People go to Pizza

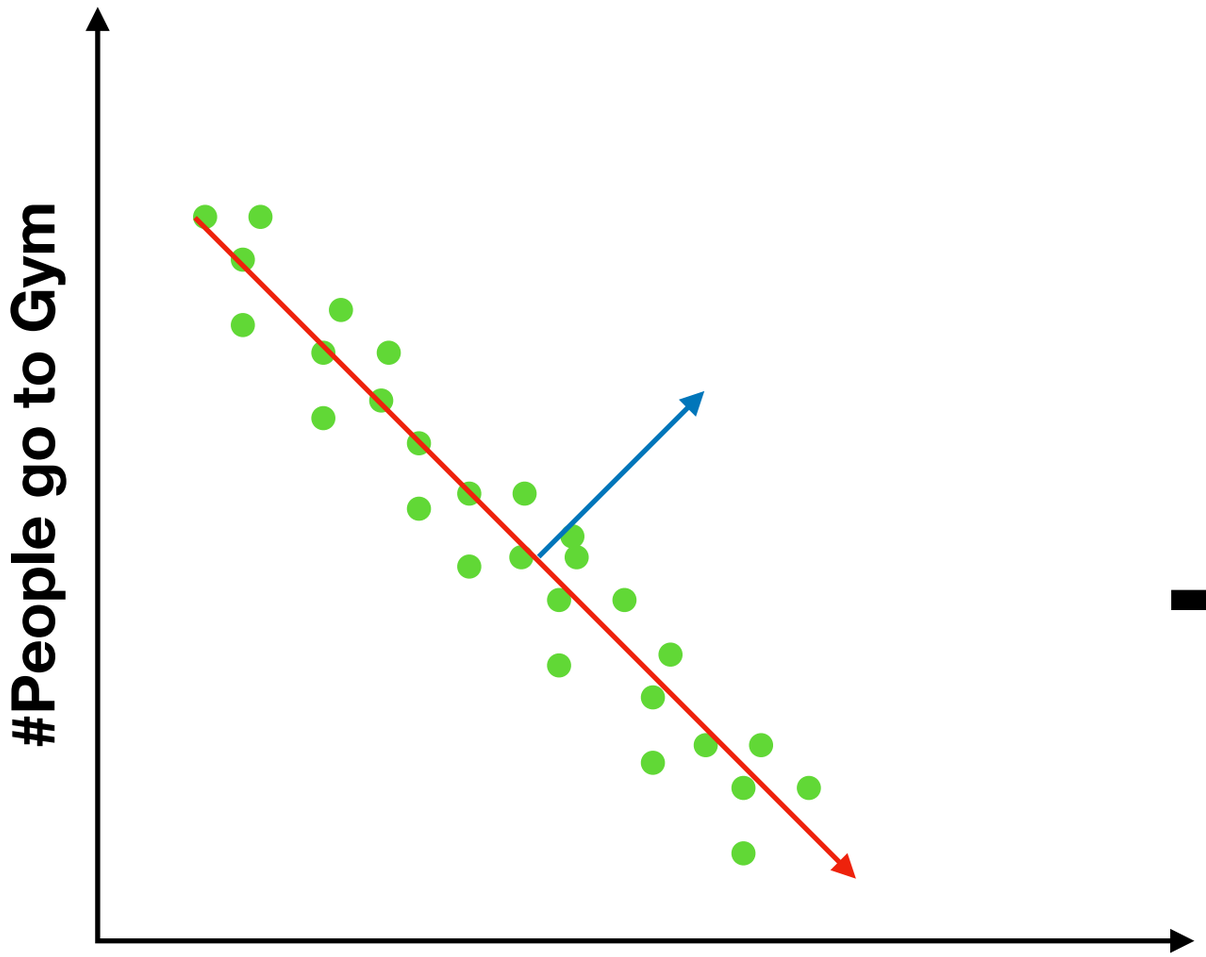


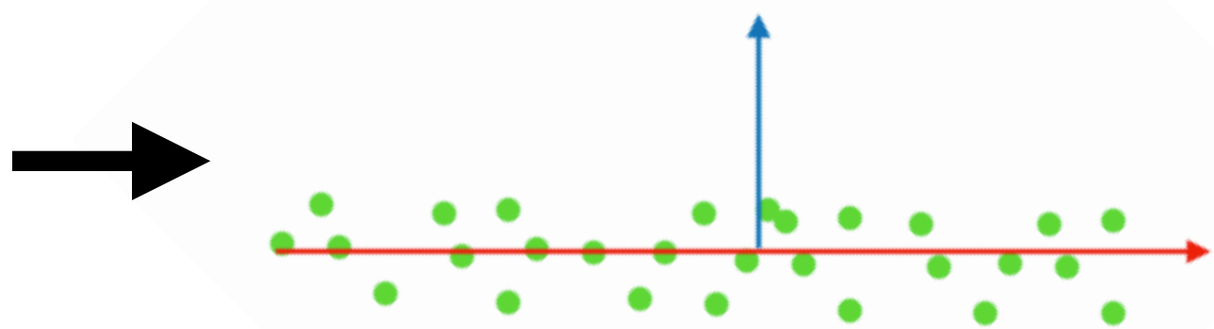
#People go to Pizza





#People go to Pizza





#People go to Pizza



PCA: Key Idea

Project centered data onto orthogonal directions that preserve as much variance as possible.

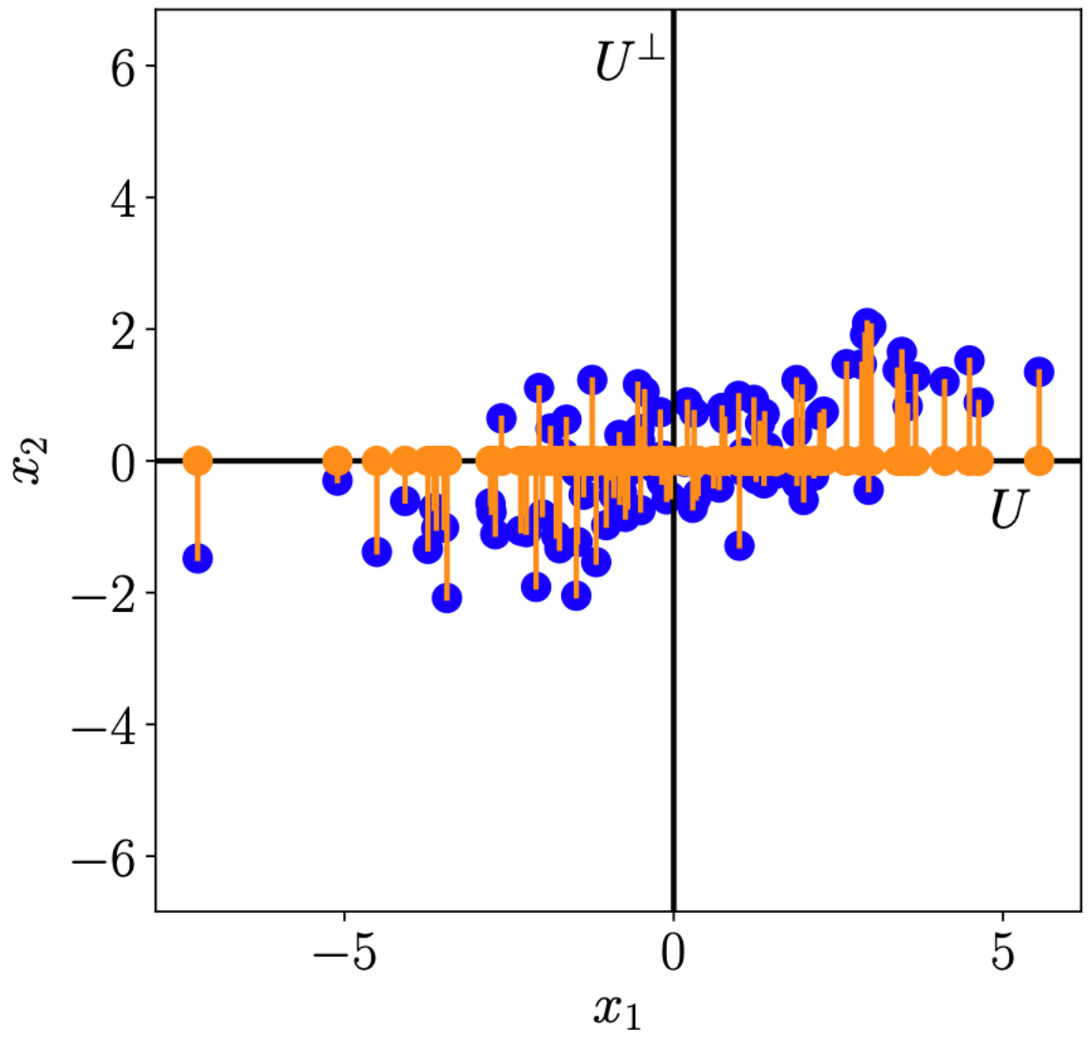
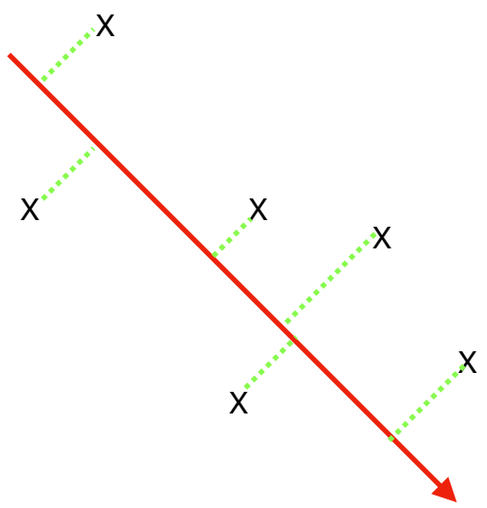


Image *from Mathematics for Machine Learning* by Deisenroth, Faisal, and Ong

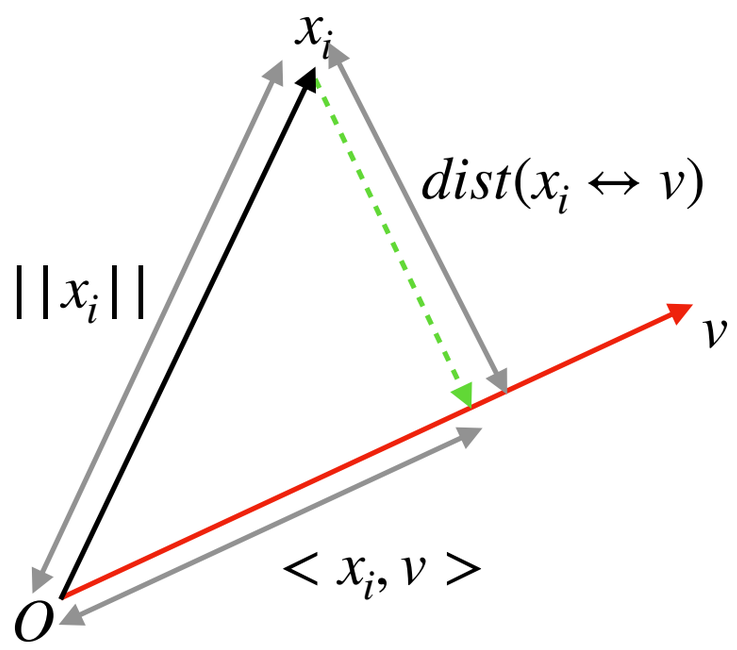
Objective Function

For centered data, PCA finds the k-dimensional subspace that minimizes the average squared reconstruction error.

$$\underset{\text{k-dim spaces } S}{\text{argmin}} \frac{1}{m} \sum_{i=1}^{m} ((\textit{distance of } x_i \textit{ from } S)^2)$$



Objective Function



$$\underset{\mathbf{v}\,:\,\|\mathbf{v}\|=1}{\operatorname{argmin}} \frac{1}{m} \sum_{i=1}^{m} \left((\text{distance between } \mathbf{x}_i \text{ and line spanned by } \mathbf{v})^2 \right)$$

$$(\text{dist}(\mathbf{x}_i \leftrightarrow \text{line}))^2 + \langle \mathbf{x}_i, \mathbf{v} \rangle^2 = \|\mathbf{x}_i\|^2$$

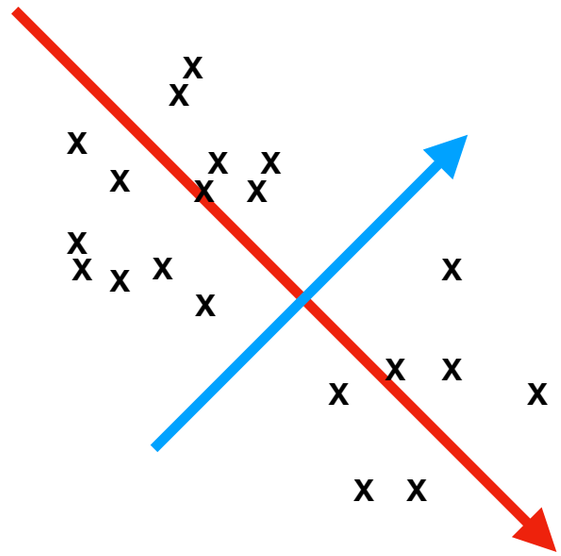
$$\underset{\mathbf{v}\,:\,\|\mathbf{v}\|=1}{\operatorname{argmax}} \frac{1}{m} \sum_{i=1}^{m} \langle \mathbf{x}_i, \mathbf{v} \rangle^2$$

Objective Function

Given $x_1, \ldots, x_m \in \mathbb{R}^n$ and a parameter $k \geq 1$, compute orthonormal vectors $v_1, \ldots, v_k \in \mathbb{R}^n$ to maximize:

$$\frac{1}{m} \sum_{i=1}^{m} \sum_{j=1}^{k} \langle x_i, v_j \rangle^2$$

The resulting k orthonormal vectors are called the top k principal components of the data.



Which is the best principal component and why?

Projection variance

$x_1, x_2, \ldots x_m \in \mathbb{R}^n$, centered observations

$$k \in \{1,2,\ldots,n\},\ \frac{1}{m}\sum_{i=1}^{m}\sum_{j=1}^{k}\langle x_i, v_j\rangle^2$$

$$X = \begin{bmatrix} -x_1- \\ -x_2- \\ \vdots \\ -x_m- \end{bmatrix}, ||v|| = 1 \rightarrow Xv = \begin{bmatrix} \langle x_1, v\rangle \\ \langle x_2, v\rangle \\ \vdots \\ \langle x_m, v\rangle \end{bmatrix}$$

Covariance quadratic form

For $k = 1$,

$$\textbf{argmax}_{v:||v||=1} \frac{1}{m} \sum_{i=1}^{m} \langle x_i, v \rangle^2 = \textbf{argmax}_{v:||v||=1} \frac{1}{m} (Xv)^\top Xv$$

$$= \textbf{argmax}_{v:||v||=1} v^\top X^\top Xv = \textbf{argmax}_{v:||v||=1} v^\top QDQ^\top v$$

By the spectral theorem we have: $X^\top X = QDQ^\top$
(Q : orthonormal eigenvectors, D : eigenvalues)

$X^\top X = (m-1)C$: same eigenvectors as the covariance matrix C, eigenvalues $(m-1)$ times larger.

Eigenvalues identify PCs

$$\textbf{argmax}_{v:||v||=1} v^\top QDQ^\top v$$

Let $w = Q^\top v$. Since Q is orthogonal, $\|w\| = \|v\| = 1$

$$v^\top QDQ^\top v = w^\top Dw = \lambda_1 w_1^2 + \lambda_2 w_2^2 + \ldots + \lambda_n w_n^2$$

Max when all weight is on $\lambda_1 \rightarrow v = v_1$

The top k PCs are the k eigenvectors of the covariance matrix with the largest eigenvalues.

Finding principal components

1. Compute the covariance matrix C of the centered data
2. Find its eigenvalues $\lambda_1 \geq \lambda_2 \geq \ldots \geq \lambda_n$ and unit eigenvectors $v_1, v_2, \ldots v_n$
3. Keep the top k : $Q_k = [v_1, v_2, \ldots, v_k]$
4. Project each centered datapoint: $x \mapsto Q_k^\top x \in \mathbb{R}^k$

PCA in Action

	Study Time(ST)	Exam Score (ES)
Student 1	10	90
Student 2	6	80
Student 3	20	100

$$D = \begin{bmatrix} 10 & 90 \\ 6 & 80 \\ 20 & 100 \end{bmatrix}, \mu = [12,90]$$

$$X = D - \mu = \begin{bmatrix} -2 & 0 \\ -6 & -10 \\ 8 & 10 \end{bmatrix}$$

$$COV = \frac{1}{m-1} X^\top X = \frac{1}{m-1} \begin{bmatrix} -2 & -6 & 8 \\ 0 & -10 & 10 \end{bmatrix} \begin{bmatrix} -2 & 0 \\ -6 & -10 \\ 8 & 10 \end{bmatrix} = \begin{bmatrix} 52 & 70 \\ 70 & 100 \end{bmatrix}$$

Two eigenvectors: $v_1 = \begin{bmatrix} 5/7 \\ 1 \end{bmatrix}$ with $\lambda_1 = 150$ and $v_2 = \begin{bmatrix} -7/5 \\ 1 \end{bmatrix}$ with $\lambda_2 = 2$

Top 1 principal component $Q_1 = \frac{v_1}{||v_1||} = \begin{bmatrix} 0.5812 \\ 0.8137 \end{bmatrix}$

Project each centered datapoint $x \in \mathbb{R}^2$ into $\mathbb{R}^1$ by $Q_1^\top x$.

Projected data into $\mathbb{R}^1$ become $[-1.1624, -11.6247, 12.7872]$

Your Guess

We replaced 2 numbers per student with 1.

How much of the variance did we keep?

A. ~50% B. ~75% C. ~90% D. ~99%

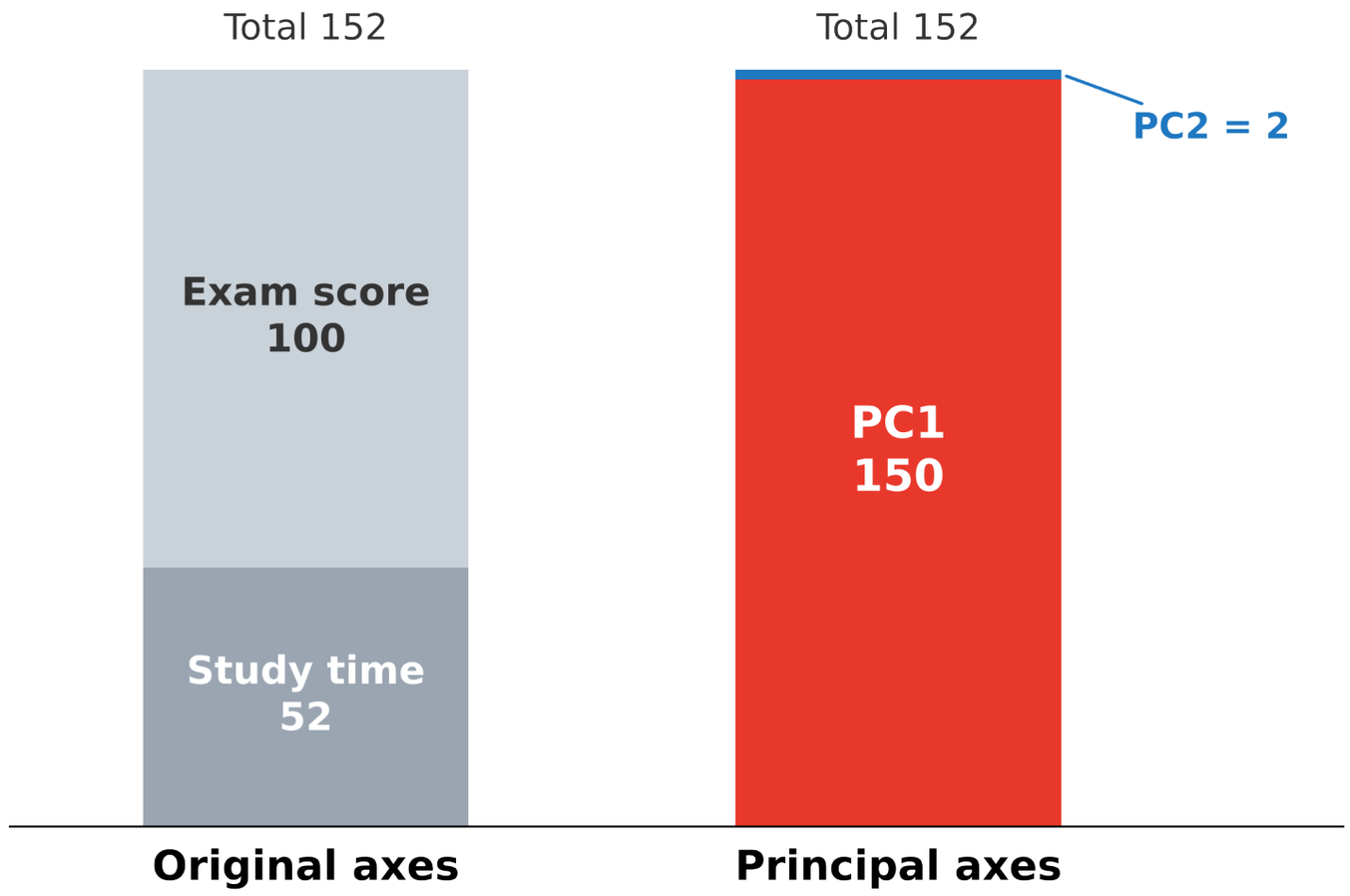
How much variance did we keep?

Explained variance of top k PCs:

$$\frac{\lambda_1 + \cdots + \lambda_k}{\lambda_1 + \cdots + \lambda_n}$$

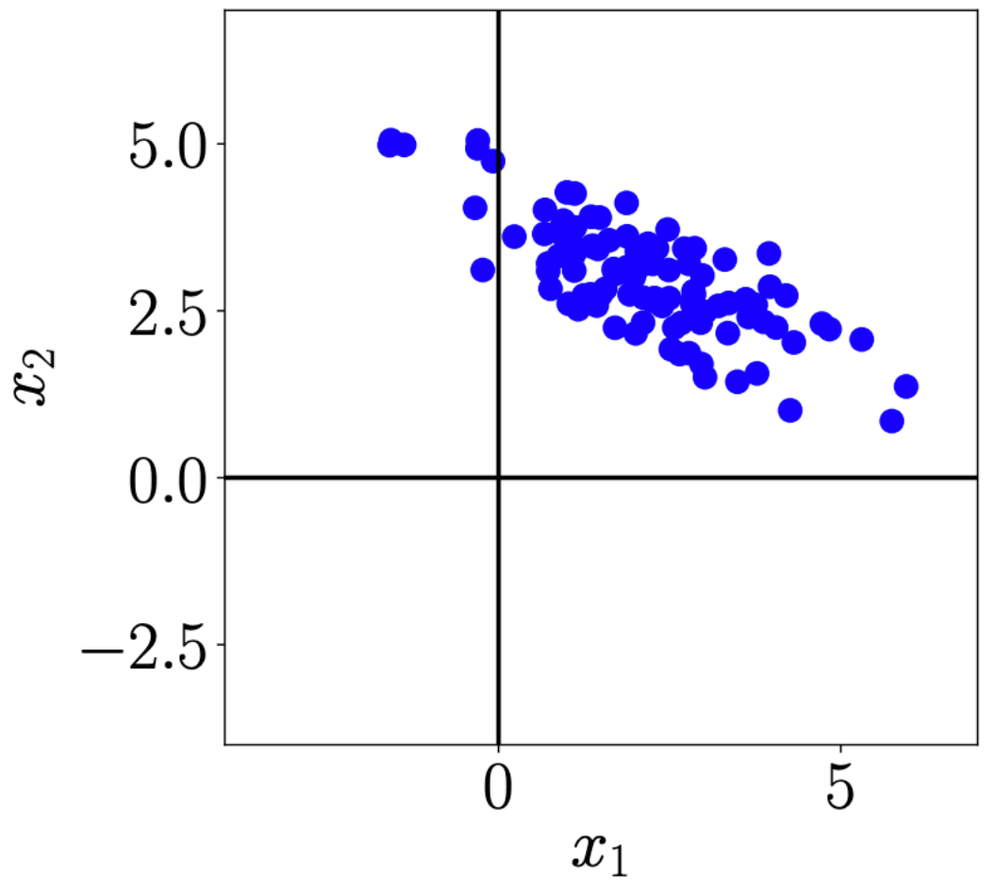
Our example:

$$\frac{150}{150+2} \approx 98.7\,\%$$



Choosing k: keep enough PCs to explain 90-95%

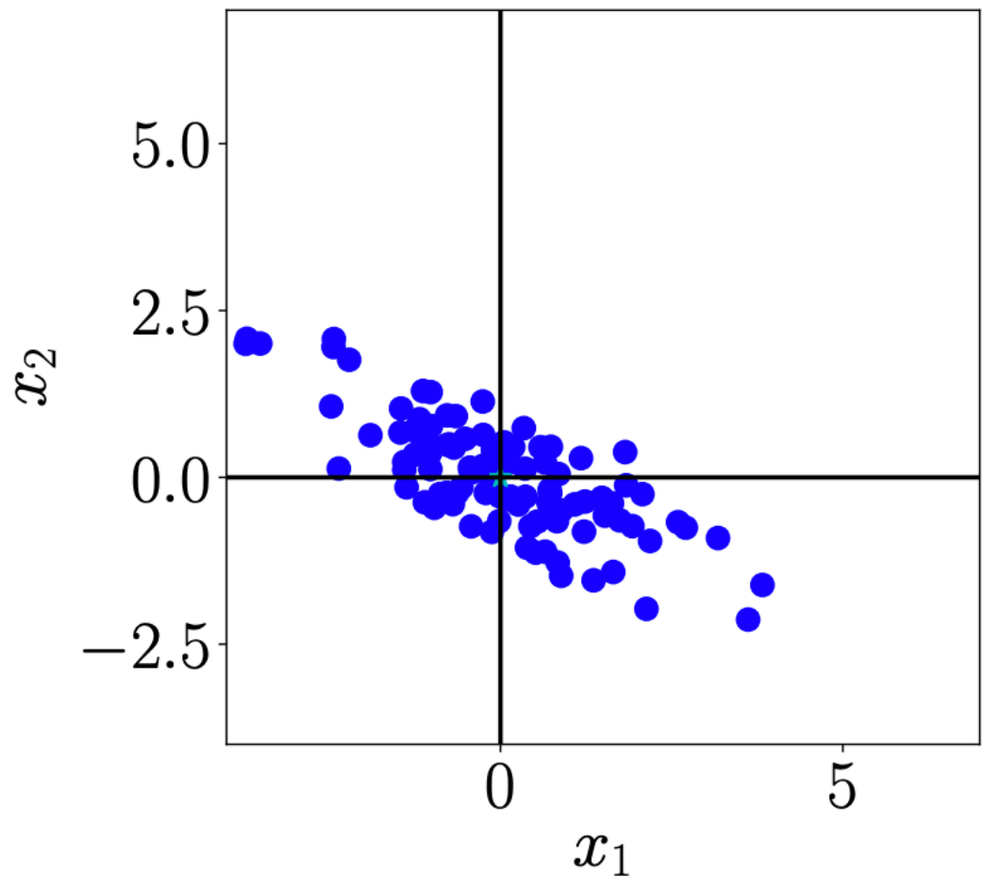
PCA in Practice



(a) Original dataset.

Image *from Mathematics for Machine Learning* by Deisenroth, Faisal, and Ong

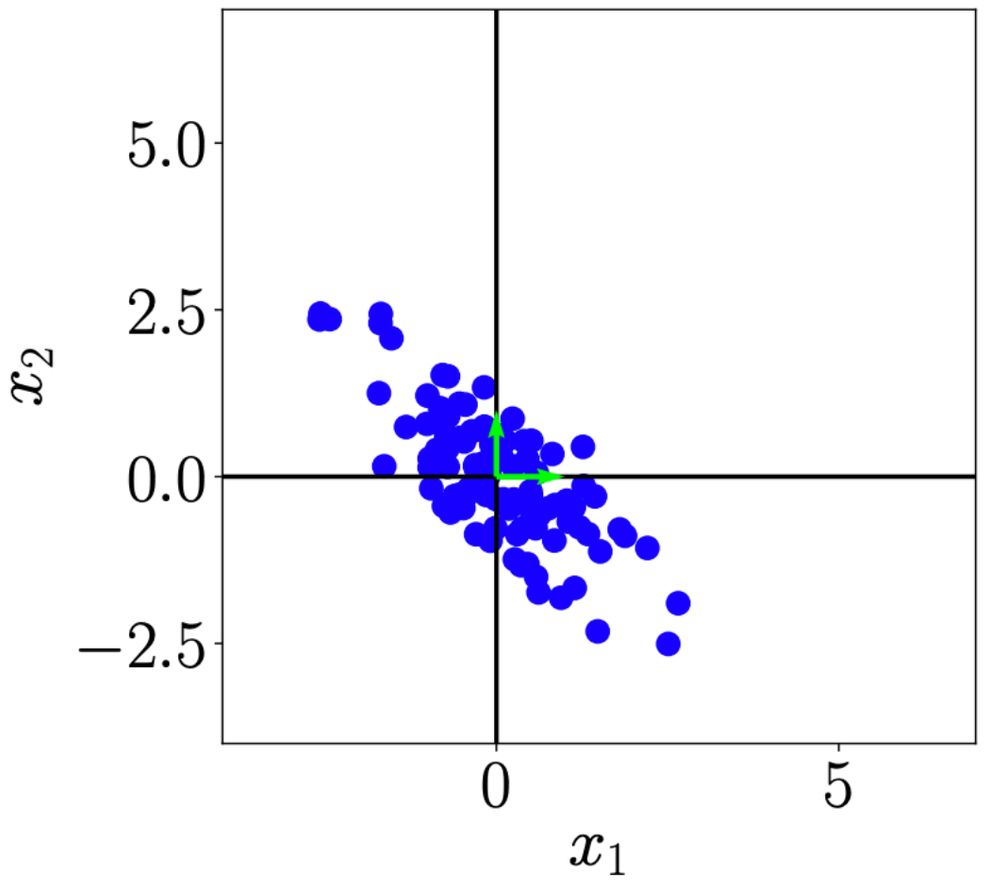
PCA in Practice



(b) Step 1: Centering by subtracting the mean from each data point.

Image *from Mathematics for Machine Learning* by Deisenroth, Faisal, and Ong

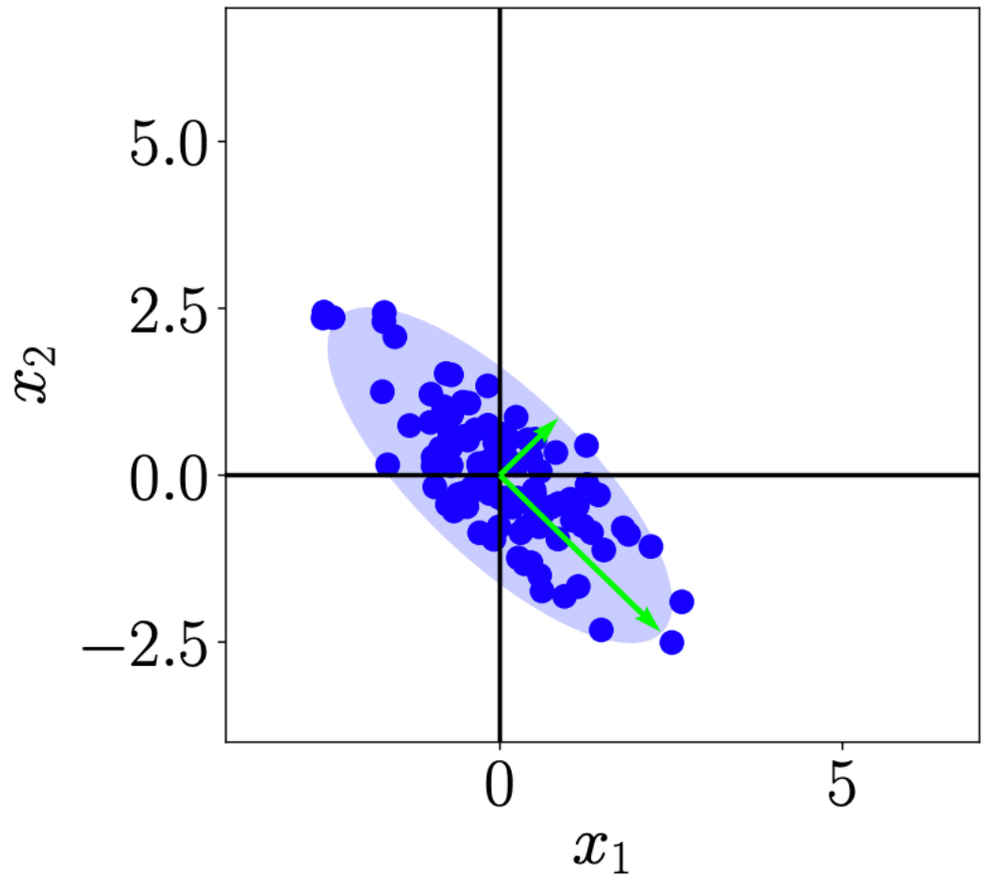
PCA in Practice



(c) Step 2: Dividing by the standard deviation to make the data unit free. Data has variance 1 along each axis.

Image *from Mathematics for Machine Learning* by Deisenroth, Faisal, and Ong

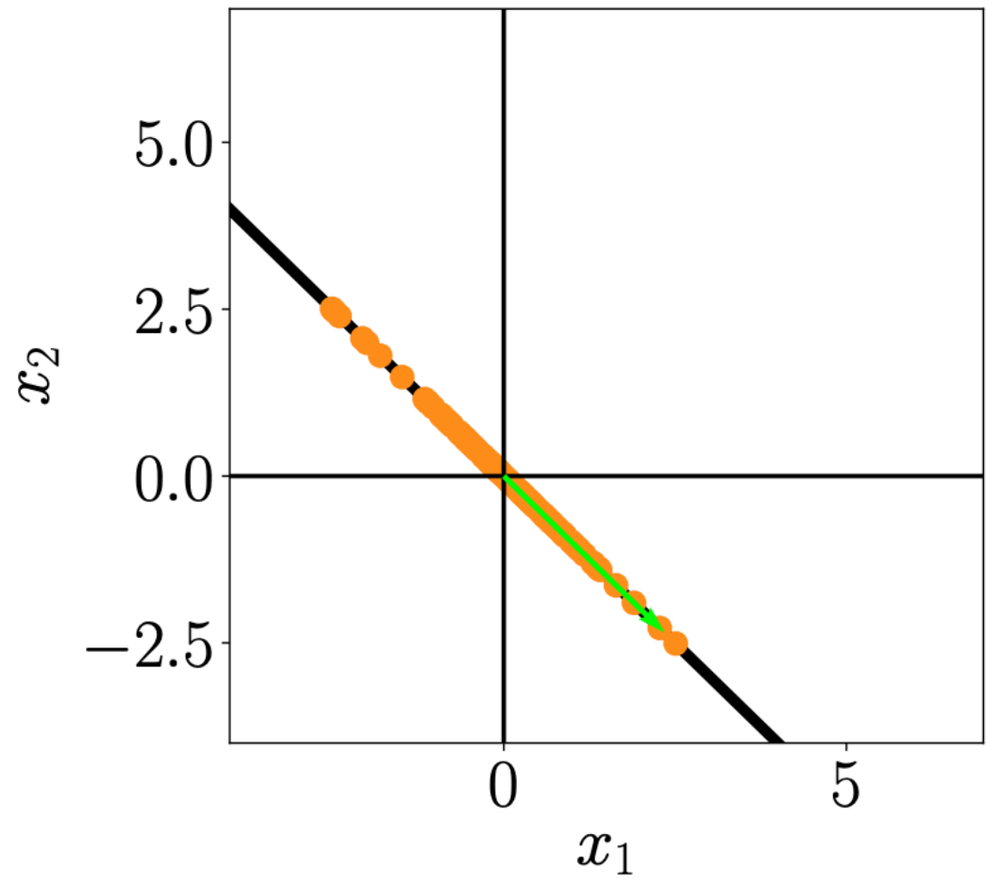
PCA in Practice



(d) Step 3: Compute eigenvalues and eigenvectors (arrows) of the data covariance matrix (ellipse).

Image *from Mathematics for Machine Learning* by Deisenroth, Faisal, and Ong

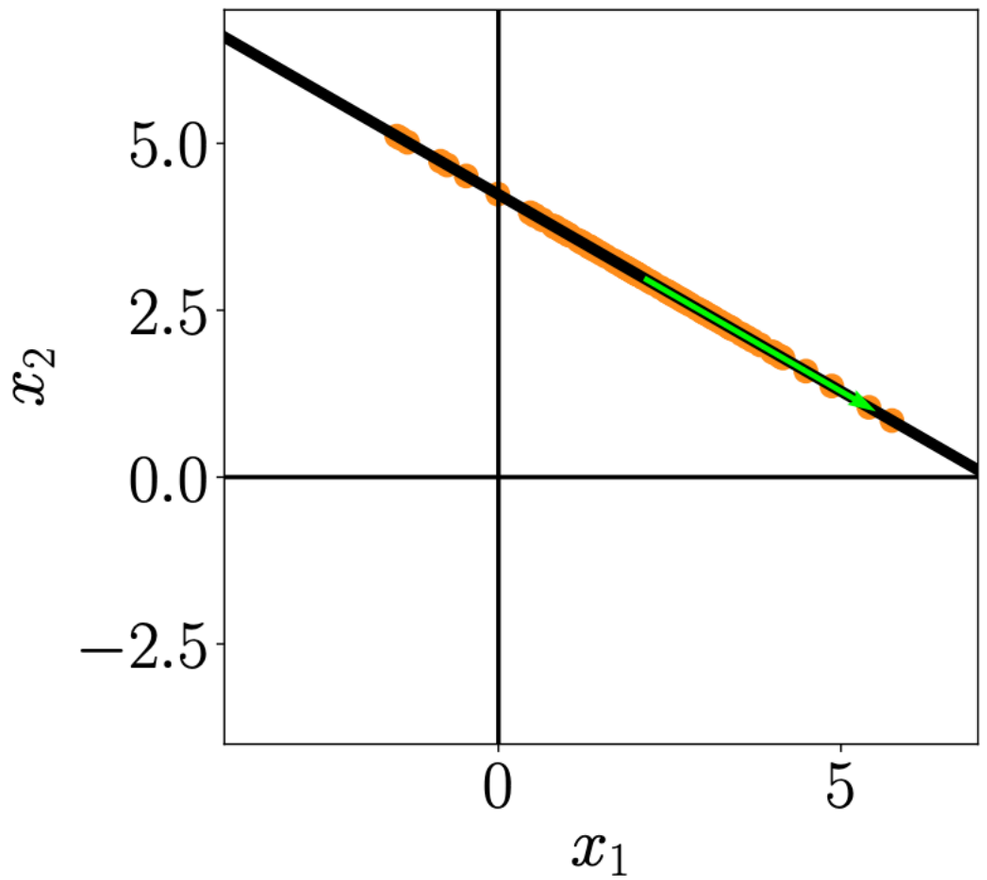
PCA in Practice



(e) Step 4: Project data onto the principal subspace.

Image *from Mathematics for Machine Learning* by Deisenroth, Faisal, and Ong

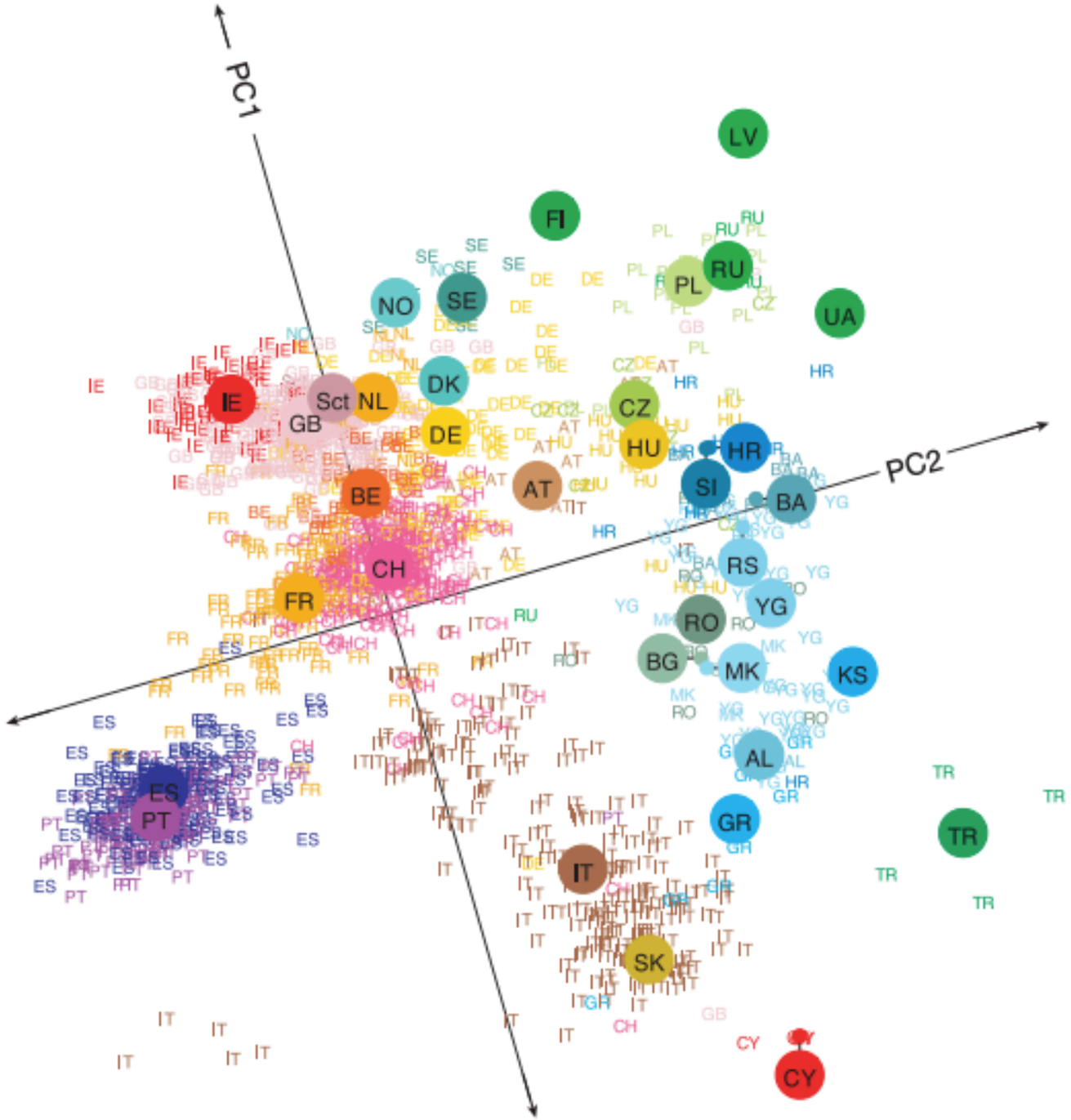
PCA in Practice



(f) Undo the standardization and move projected data back into the original data space from (a).

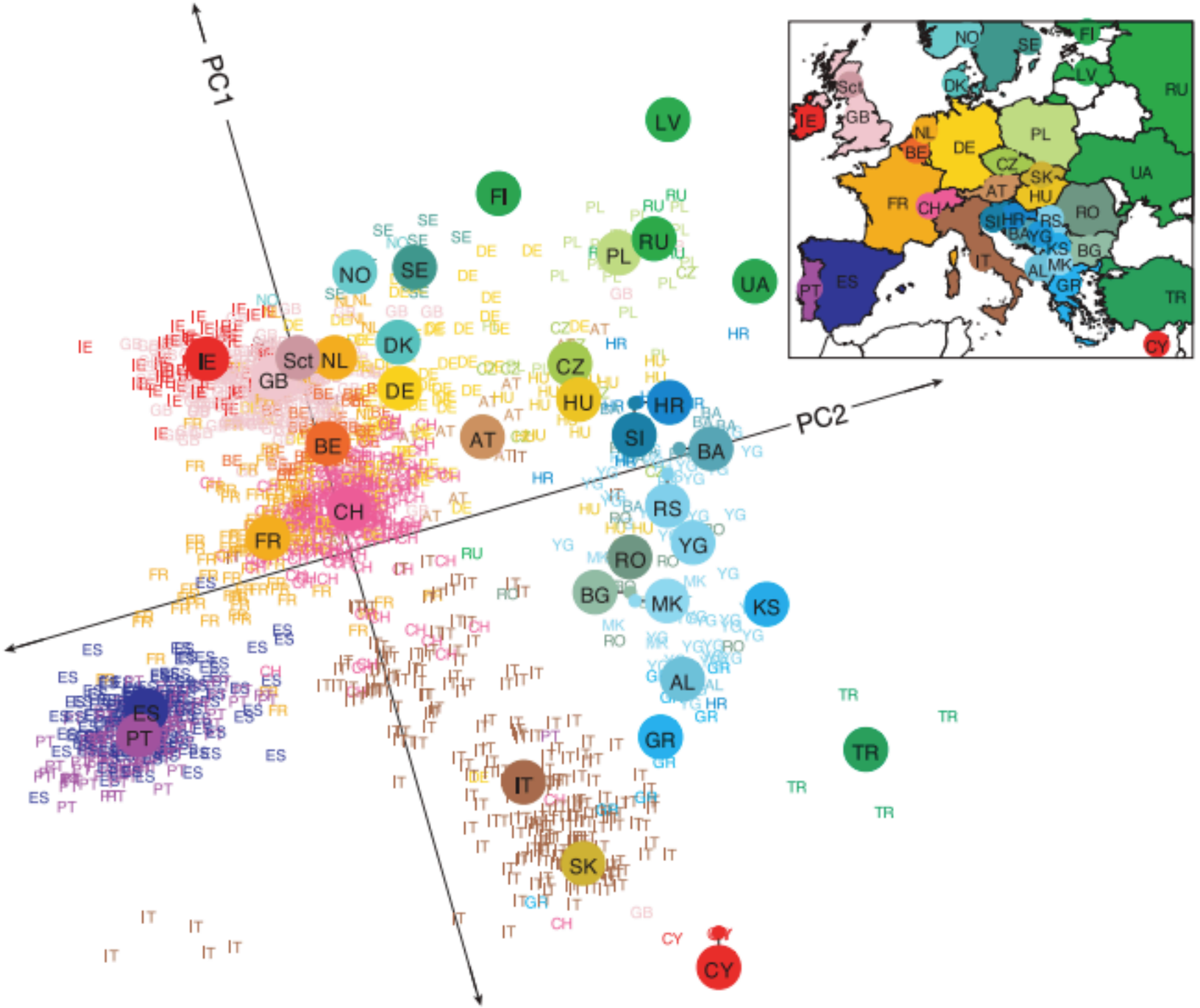
Image *from Mathematics for Machine Learning* by Deisenroth, Faisal, and Ong

What do you see?



Novembre, John, et al. "Genes mirror geography within Europe." *Nature* 456.7218 (2008): 98-101.

PCA finds Europe in our DNA



Novembre, John, et al. "Genes mirror geography within Europe." *Nature* 456.7218 (2008): 98-101.

Faces as a sum of “ghost faces”

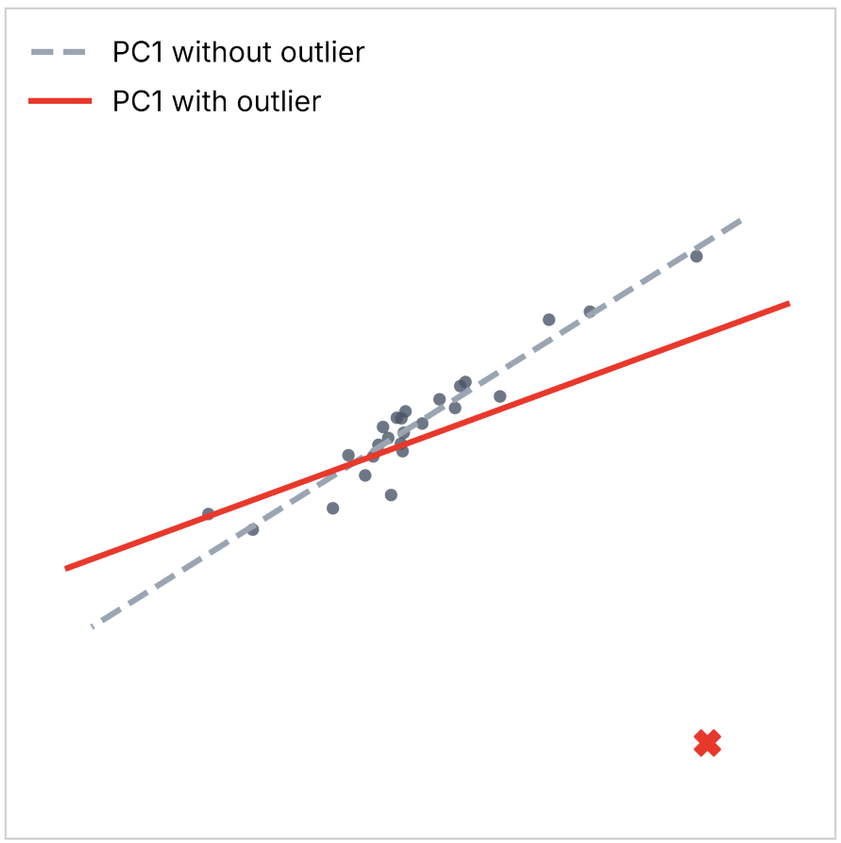


- Each image = one point in $\mathbb{R}^{\text{pixels}}$
- Eigenfaces = principal components of many faces
- Store a face as a few numbers $(w_1, w_2, \ldots)$
- Remember slide 2? Each face here is ~600 pixels, stored with just a handful of numbers.

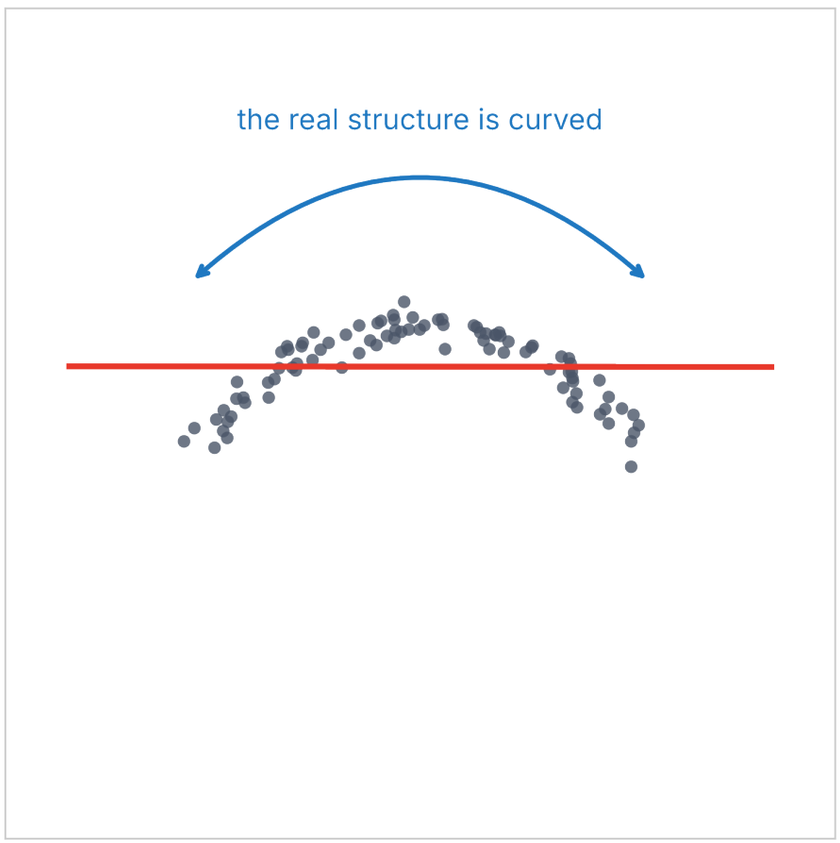
Turk, Matthew, and Alex Pentland. "Eigenfaces for recognition." *Journal of cognitive neuroscience* 3.1 (1991): 71-86.
Eigenfaces computed from Labeled Faces in the Wild (Huang et al., 2007)

Failure Cases

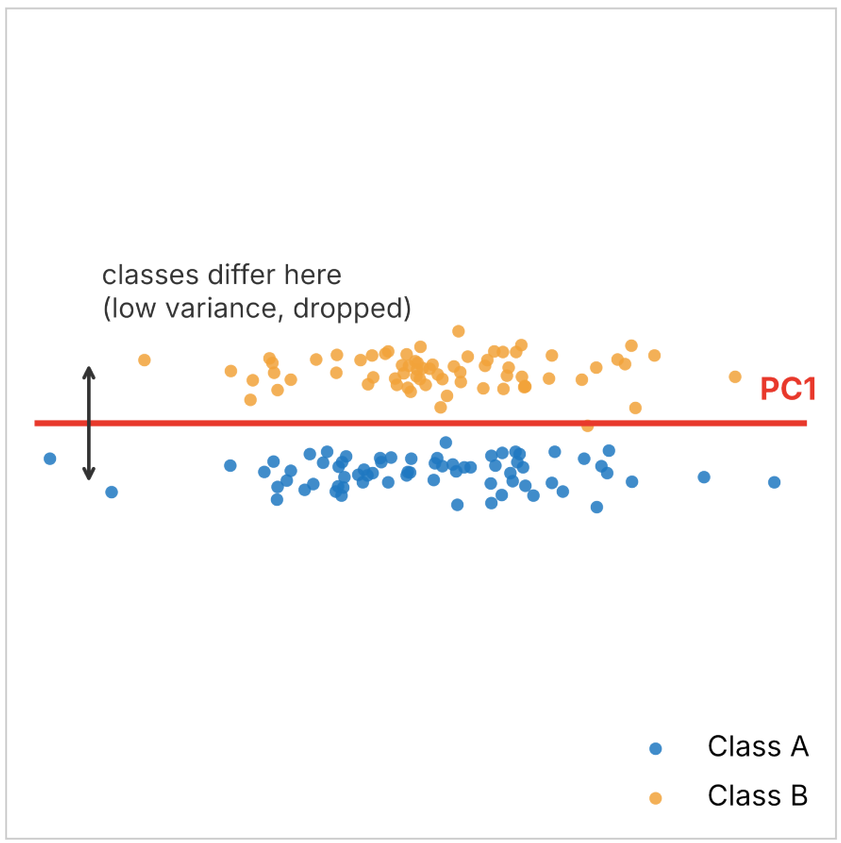
One outlier tilts PC1



A straight line misses curves



Useful signal can be low-variance



Key Takeaways

- PCA finds the directions where the data varies most
- Principal components = top eigenvectors of the covariance matrix
- Eigenvalues tell how much variance each component keeps
- Center (and usually scale) the data first
- PCA is linear and sensitive to outliers

Thank you

Questions?

References

- https://web.stanford.edu/class/cs168/l/l7.pdf
- https://web.stanford.edu/class/cs168/l/l8.pdf
- https://www.youtube.com/watch?v=g-Hb26agBFg&t=1121s
- Turk, Matthew, and Alex Pentland. "Eigenfaces for recognition." *Journal of cognitive neuroscience* 3.1 (1991): 71-86.
- Novembre, John, et al. "Genes mirror geography within Europe." *Nature* 456.7218 (2008): 98-101.
- Huang, G. B., Ramesh, M., Berg, T., & Learned-Miller, E. (2007). Labeled Faces in the Wild: A database for studying face recognition in unconstrained environments. University of Massachusetts Amherst, Technical Report 07-49.
- Deisenroth, Marc Peter, A. Aldo Faisal, and Cheng Soon Ong. *Mathematics for Machine Learning.* Cambridge University Press, 2020.